

第二章 量子力学导论

一、波粒二象性

1.

v_1, v_2 两波叠加 $\rightarrow$ 拍

$\Delta v = |v_1 - v_2|$ 已知 v_1, v_2 , 至少观察一个拍

$$\Delta t \gg \frac{1}{\Delta v} \rightarrow \Delta t \Delta v \gg 1$$

$$v = \frac{v}{\lambda} \rightarrow \Delta x \Delta \lambda \gg \lambda^2$$

$$\Delta v = \frac{v}{\lambda} \Delta \lambda, \Delta x = v \Delta t$$

2.

$$E = h\nu \quad p = \frac{h}{\lambda} = \hbar k$$

3.

电子稳定绕核运动 $\rightarrow$ 形成驻波

$$2\pi r = n\lambda = n \frac{h}{p}$$

$$\hookrightarrow L = pr = n\hbar$$

刚性匣中粒子:

$$E_n = \frac{1}{2} m v^2, \quad T = \frac{2a}{v}$$

驻波条件: $n \cdot \frac{a}{2} = a$

$$\hookrightarrow p = \frac{h}{\lambda} = \frac{n\hbar}{2a}$$

$$E_n = \frac{p^2}{2m} = \frac{n^2 \hbar^2}{8ma^2}$$

动量、能量均量子化

应用到氢原子模型: $2a = 2\pi r$

$$E_n = \frac{\hbar^2}{8m\pi^2 r^2} \quad (\text{假设处于基态})$$

$$\hbar = 2\pi\hbar$$

$$E = E_n + E_p = \frac{\hbar^2}{8m\pi^2 r^2} - \frac{e^2}{4\pi\epsilon_0 r}$$

$$\frac{dE}{dr} \Big|_{r=r_{min}} = -\frac{\hbar^2}{4m\pi^2 r_{min}^3} + \frac{e^2}{4\pi\epsilon_0 r_{min}^2} = 0 \rightarrow r_{min} = \frac{4\pi\epsilon_0 \hbar^2}{me^2} = a_1$$

二、不确定关系

粒子在客观上不能同时具有确定的坐标位置和相应动量

$$\Delta x = \frac{h}{\Delta p_x}$$

$$\hookrightarrow \Delta x \Delta p_x = \frac{h}{\Delta p_x} \Delta p_x \Delta x \geq h$$

$$\Rightarrow \Delta x \Delta p_x \geq h$$

$$\Delta v = \frac{1}{h} \Delta E$$

$$\hookrightarrow \Delta t \Delta v = \frac{1}{h} \Delta t \Delta E \geq h$$

$$\Rightarrow \Delta t \Delta E \geq h$$

例1:

假设电子被束缚在 r 的范围内

$$\Delta p_x = \frac{h}{\Delta x} = \frac{h}{2r}$$

$$\overline{p_x^2} = \epsilon p_x^2 = \frac{1}{2} \overline{p^2}$$

$$\overline{E} = \frac{\overline{p^2}}{2m} = \frac{h^2}{8m r^2}$$

§ 14 波函数

$$\psi = \psi_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$$

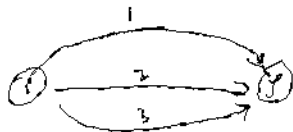
$$N \propto |\psi|^2$$

态的叠加:

从态 $i \rightarrow j$ 的跃迁概率

$$W_{ij} = |\langle f|i \rangle|^2$$

$$(1) \langle f|i \rangle = \sum_n \langle f|n \rangle \langle n|i \rangle$$

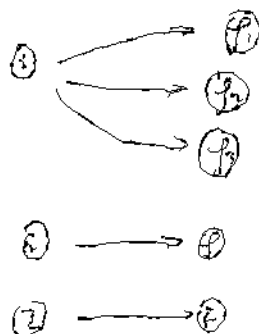


$$(2) |\langle f|i \rangle|^2 = \left| \sum_n \langle f|n \rangle \langle n|i \rangle \right|^2$$

$$(3) \langle f|i \rangle = \langle f|U \rangle \langle U|i \rangle$$



$$(4) \langle f|U^\dagger|i \rangle = \langle f|i \rangle \langle f|U \rangle$$



干涉实验:

只开缝 1

只开缝 2

$$\langle x|S \rangle_1 = \langle x|1 \rangle \langle 1|S \rangle$$

$$\langle x|S \rangle_2 = \langle x|2 \rangle \langle 2|S \rangle$$

1, 2 一起开

$$\langle x|S \rangle = \langle x|1 \rangle \langle 1|S \rangle + \langle x|2 \rangle \langle 2|S \rangle$$

$$I \propto |\langle x|S \rangle|^2 = I_0 + I_0 + \langle x|S \rangle^* \langle x|S \rangle_1 + \langle x|S \rangle_2^* \langle x|S \rangle_1$$

$$= I_0 + I_0 + \boxed{I_{\text{干涉项}}}$$

利用光子来观察时:

$$\psi_1 = \langle x|1 \rangle \langle 1|S \rangle \text{ 电子通过1到 } x$$

$$\psi_2 = \langle x|2 \rangle \langle 2|S \rangle \text{ 电子通过2到 } x$$

$$\psi_1 = \langle D_1|1 \rangle \langle 1|P \rangle = \langle D_2|2 \rangle \langle 2|P \rangle$$

光子在1处被探测到 D_1 (对称性)

$$\psi_2 = \langle D_1|2 \rangle \langle 2|P \rangle = \langle D_2|1 \rangle \langle 1|P \rangle$$



电子到达 x , 光子在 D_1 被记录:

$$\begin{aligned}\langle x D_1 | SP \rangle &= \langle x | 1 \rangle \langle 1 | S \rangle \langle D_1 | 1 \rangle \langle 1 | P \rangle \\ &+ \langle x | 2 \rangle \langle 2 | S \rangle \langle D_1 | 2 \rangle \langle 2 | P \rangle \\ &= \varphi_1 \psi_1 + \varphi_2 \psi_2\end{aligned}$$

同理: $\langle x D_2 | SP \rangle = \varphi_2 \psi_1 + \varphi_1 \psi_2$

$$\begin{aligned}Rq |\langle x | S \rangle|^2 &= |\langle x D_1 | SP \rangle|^2 + |\langle x D_2 | SP \rangle|^2 \\ &= (\varphi_1 \psi_1 + \varphi_2 \psi_2)(\varphi_1^* \psi_1^* + \varphi_2^* \psi_2^*) + (\varphi_2 \psi_1 + \varphi_1 \psi_2)(\varphi_2^* \psi_1^* + \varphi_1^* \psi_2^*) \\ &= \varphi_1^2 \psi_1^2 + \varphi_2^2 \psi_2^2 + \varphi_2^2 \psi_1^2 + \varphi_1^2 \psi_2^2 + \varphi_1 \varphi_2^* \psi_1 \psi_2^* + \varphi_1^* \varphi_2 \psi_1^* \psi_2 \\ &\quad + \varphi_1^* \varphi_2 \psi_1 \psi_2^* + \varphi_1 \varphi_2^* \psi_1^* \psi_2 \\ &= (\varphi_1^2 + \varphi_2^2)(\psi_1^2 + \psi_2^2) + \underbrace{(\varphi_1 \varphi_2^* + \varphi_1^* \varphi_2)(\psi_1 \psi_2^* + \psi_1^* \psi_2)}_{\text{干涉项}}\end{aligned}$$

要确认电子从那条路径走

§ 15

1. 一维粒 δ :

$$E = \frac{p^2}{2m} + V(x) \quad E = \hbar\omega, \quad p = \hbar k$$

$$\hbar\omega = \frac{\hbar^2 k^2}{2m} + V(x)$$

平面波: $\psi = \psi_0 e^{i(kx - \omega t)}$

$$\begin{cases} i\hbar \frac{\partial}{\partial t} \psi = \hbar\omega \psi = E\psi \\ -i\hbar \frac{\partial}{\partial x} \psi = \hbar k \psi = p\psi \\ -\hbar^2 \frac{\partial^2}{\partial x^2} \psi = (\hbar k)^2 \psi = p^2 \psi \end{cases}$$

$$\Rightarrow i\hbar \frac{\partial}{\partial t} \psi = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi + V(x) \psi$$

三维情况: $[-\frac{\hbar^2}{2m} \nabla^2 \psi + V(\vec{r})] \psi(\vec{r}, t) = i\hbar \frac{\partial}{\partial t} \psi$

其中 $\psi(\vec{r}, t) = \psi_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$

叠加原理: ψ_1, ψ_2 为解, $C_1 \psi_1 + C_2 \psi_2$ 为解

2. 定态薛定谔方程 ($V(\vec{r})$ 不是含 t)

$$\psi(\vec{r}, t) = \psi(\vec{r}) T(t)$$

$$\frac{1}{\psi} [-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r})] \psi = \frac{1}{T} i\hbar \frac{\partial T}{\partial t} = E$$

$$\rightarrow \frac{dT}{T} = \frac{E}{i\hbar} dt \quad \text{解得} \quad T = e^{-\frac{iE}{\hbar} t}$$

$$\therefore \bar{\psi} = \psi(\vec{r}) e^{-\frac{iEt}{\hbar}}$$

$$[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r})] \psi = E \psi$$

例：一维无限深势阱

$$V(x) = \begin{cases} 0 & 0 < x < d \\ \infty & x > d, x < 0 \end{cases}$$

定态薛定谔方程： $\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + E \psi = 0$

解得 $\psi(x) = A \sin(\sqrt{\frac{2mE}{\hbar^2}} x + \varphi)$

边界条件： $\psi(0) = \psi(d) = 0$ 另外 $V(x) = \infty$, $\psi(x)$ 必须为 0, 否则 $V(x)\psi$ 发散

$\hookrightarrow \sqrt{\frac{2mE}{\hbar^2}} d = n\pi \Rightarrow \psi(x) = A \sin(\frac{n\pi x}{d})$

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2m d^2}$$

归一化条件： $\int_0^d \psi^2(x) dx = 1$

$$A^2 \int_0^d \sin^2(\frac{n\pi x}{d}) dx = \frac{1}{2} A^2 d \Rightarrow A = \sqrt{\frac{2}{d}}$$

$$\psi(x) = \sqrt{\frac{2}{d}} \sin(\frac{n\pi x}{d})$$

例：一维有限深势阱

$$V(x) = \begin{cases} 0 & x \leq \frac{d}{2} \text{ 或 } x \geq \frac{d}{2} \\ V_0 & |x| < \frac{d}{2} \end{cases}$$

在势阱内： $\psi(x) = \sqrt{\frac{2}{d}} \sin(\frac{n\pi x}{d})$

在势阱外： $-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V_0 \psi = E \psi$

$$\hookrightarrow \frac{d^2 \psi}{dx^2} = \frac{2m(V_0 - E)}{\hbar^2} \psi, \quad k = \frac{1}{\hbar} \sqrt{2m(V_0 - E)}$$

边界条件： $\psi(0) = 0$ / $\psi(d) = 0$

通解： $\psi(x) = A e^{-kx} + B e^{kx}$

§ 16 平均值和算符

$$\overline{f(x)} = \int_{-\infty}^{\infty} \psi^*(x) f(x) \psi(x) dx$$

$$\text{其中 } \int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx = 1$$

不确定关系：\$P\$ 的没有意义，只能在动量空间中求 \$\bar{p}\$

$$\text{傅里叶变换: } \phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x) e^{-ikx} dx \quad k = \frac{p}{\hbar}$$

为 \$\frac{1}{\sqrt{2\pi}}\$?

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ikx} dk$$

动量 \$P_x\$ 在 \$p_x \sim p_x + dp_x\$ 区间概率为 \$\phi^*(k) \phi(k) dk\$

$$\text{则 } \bar{p}_x = \int_{-\infty}^{\infty} \phi^*(k) p_x \phi(k) dk$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{\infty} \psi^*(x') e^{ikx'} dx' \right) \hbar k \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{\infty} \psi(x) e^{-ikx} dx \right) dk$$

$$= \frac{i\hbar}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \psi^*(x') e^{ikx'} dx' \right) \left(\int_{-\infty}^{\infty} \psi \frac{\partial}{\partial x} (e^{-ikx}) dx \right) dk$$

$$= \frac{i\hbar}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \psi^*(x') e^{ikx'} dx' \right) \left(\psi(x) e^{-ikx} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{\partial \psi(x)}{\partial x} e^{-ikx} dx \right) dk$$

$$\psi(\infty) = \psi(-\infty) = 0$$

$$\frac{\partial}{\partial k} \int_{-\infty}^{\infty} f(x) dx \rightarrow \int_{-\infty}^{\infty} f(x) \frac{\partial}{\partial k} dx$$

$$= -i\hbar \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi^*(x') e^{ikx'} dx' \right) e^{-ikx} dx \right] \frac{\partial \psi(x)}{\partial x} dx$$

$$\underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi^*(x') e^{ikx'} dx' \right) e^{-ikx} dx}_{= \phi^*(k)} = \psi^*(x)$$

$$= \int_{-\infty}^{\infty} \psi^* (-i\hbar \frac{\partial}{\partial x}) \psi dx$$

其中 \$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}\$ 称为动量算符

$$\hookrightarrow \hat{p} = -i\hbar \nabla, \quad \hat{E} = i\hbar \frac{\partial}{\partial t}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\hat{L}_x = \hat{y} \hat{p}_z - \hat{z} \hat{p}_y = -i\hbar (y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y})$$

$$\hat{L}_y = \hat{x} \hat{p}_z - \hat{z} \hat{p}_x = -i\hbar (x \frac{\partial}{\partial z} - z \frac{\partial}{\partial x})$$

$$\hat{L}_z = \hat{x} \hat{p}_y - \hat{y} \hat{p}_x = -i\hbar (x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x})$$

球坐标:

$$\hat{L}_x = -i\hbar \left[r \sin\theta \sin\varphi \left(\frac{\partial}{\partial r} \frac{\partial}{\partial x} + \frac{\partial}{\partial \theta} \frac{\partial}{\partial x} + \frac{\partial}{\partial \varphi} \frac{\partial}{\partial x} \right) + r \sin\theta \left(\frac{\partial}{\partial r} \frac{\partial}{\partial y} + \frac{\partial}{\partial \theta} \frac{\partial}{\partial y} + \frac{\partial}{\partial \varphi} \frac{\partial}{\partial y} \right) \right]$$

$$\hat{L}_x = i\hbar \left(\sin\phi \frac{\partial}{\partial \theta} + \cot\theta \cos\phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_y = -i\hbar \left(\cos\phi \frac{\partial}{\partial \theta} - \cot\theta \sin\phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial \phi^2} \right]$$

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y$$

$$[\hat{L}^2, \hat{L}] = 0$$

本征方程, 本征函数, 本征值.

$$\text{算符 } \hat{O} f = y = \lambda f$$

f 称为本征函数, λ 为本征值

粒子在三维空间中的力学量 $\hat{A}$ 的平均值为

$$\bar{A} = \int \psi^*(\vec{r}) \hat{A} \psi(\vec{r}) d^3r$$

对易:

$$[\hat{G}, \hat{F}] = \hat{G}\hat{F} - \hat{F}\hat{G} = 0 \quad \text{对易} \\ \neq 0 \quad \text{不对易}$$

$$\begin{aligned} [\hat{x}, \hat{p}_x] \psi &= (\hat{x} \hat{p}_x - \hat{p}_x \hat{x}) \psi \\ &= x (-i\hbar \frac{\partial}{\partial x} \psi) - [-i\hbar \frac{\partial}{\partial x} (x \psi)] \\ &= x p_x \psi + i\hbar \psi - x p_x \psi = i\hbar \psi \end{aligned}$$

$$\Rightarrow [\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = [\hat{z}, \hat{p}_z] = i\hbar$$

$$\begin{aligned} [\hat{x}, \hat{p}_y] \psi &= (\hat{x} \hat{p}_y - \hat{p}_y \hat{x}) \psi \\ &= x (-i\hbar \frac{\partial}{\partial y} \psi) - (-i\hbar \frac{\partial}{\partial y} (x \psi)) \\ &= x p_y \psi - x p_y \psi = 0 \end{aligned}$$

$$\Rightarrow [\hat{r}_i, \hat{p}_j] = i\hbar \delta_{ij}$$

哈密顿算符: $H = -\frac{\hbar^2}{2m} \nabla^2 + V(r)$

∴ 定态薛定谔方程: $H\psi = E\psi$

§ 17 氢原子的薛定谔方程

一、中心力场的薛定谔方程

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

$$\left[-\frac{\hbar^2}{2m_\mu} \nabla^2 + V(r) \right] \psi = E\psi \quad \text{其中 } m_\mu = \frac{m_e m_p}{m_e + m_p}$$

$V(r)$ 是中心对称势 $\rightarrow$ 采用球坐标

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2}{\partial \phi^2}$$

$$\hookrightarrow -\frac{\hbar^2}{2m_\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 \psi}{\partial \phi^2} \right] + V(r)\psi = E\psi$$

$\propto \hat{L}^2$

$$-\frac{\hbar^2}{2m_\mu} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{\hat{L}^2}{2m_\mu r^2} \psi + V(r)\psi = E\psi$$

有效势

分离变量法 $\psi(r, \theta, \phi) = R(r) Y(\theta, \phi)$

$$R(r) \left\{ -\frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial R(r)}{\partial r} \right] - \frac{2m_\mu}{\hbar^2} [E - V(r)] R(r) \right\} = \frac{\hat{L}^2 Y(\theta, \phi)}{\hbar^2 Y(\theta, \phi)}$$

LHS 只与 r 相关, RHS 只与 θ, ϕ 相关 $\rightarrow$ LHS = RHS = 常数

$$\Rightarrow \begin{cases} \hat{L}^2 Y(\theta, \phi) = \alpha \hbar^2 Y(\theta, \phi) & \textcircled{1} \\ -\frac{1}{R(r)} \frac{\partial}{\partial r} \left[r^2 \frac{\partial R(r)}{\partial r} \right] - \frac{2m_\mu r^2}{\hbar^2} [E - V(r)] = \alpha & \textcircled{2} \end{cases}$$

$$\textcircled{1} = Y(\theta, \phi) = \Theta(\theta) \Phi(\phi)$$

$$-\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial \phi^2} \right] \Theta(\theta) \Phi(\phi) = \alpha^2 \hbar^2 \Theta(\theta) \Phi(\phi)$$

$$-\frac{1}{\Theta(\theta)} \frac{1}{\sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta(\theta)}{\partial \theta} \right) - \frac{1}{\Phi(\phi)} \frac{1}{\sin^2\theta} \frac{\partial^2 \Phi(\phi)}{\partial \phi^2} = \alpha^2$$

$$\frac{\sin\theta}{\Theta(\theta)} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta(\theta)}{\partial \theta} \right) + \alpha^2 \sin^2\theta = -\frac{1}{\Phi(\phi)} \frac{\partial^2 \Phi(\phi)}{\partial \phi^2} = \beta$$

$$\Rightarrow \left\{ \begin{aligned} \frac{\partial^2 \psi}{\partial \varphi^2} &= -\psi \quad (3) \\ \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial \psi}{\partial \theta}) &= -\alpha^2 \sin \theta \psi \quad (4) \end{aligned} \right.$$

$$(3): \text{解得 } \psi = \begin{cases} A e^{iN\varphi} + B e^{-iN\varphi} & \nabla \neq 0 \\ C + D\varphi & \nabla = 0 \end{cases}$$

波函数在端点单值性: $\psi(\varphi) = \psi(\varphi + 2\pi)$

$$\text{和归一化条件: } \int_0^{2\pi} \psi(\varphi) \psi^*(\varphi) d\varphi = 1$$

$$\hookrightarrow \psi_m = \frac{1}{\sqrt{2\pi}} e^{im\varphi} \quad m = 0, \pm 1, \dots$$

$$\begin{aligned} \hat{L}_z \psi_m &= -i\hbar \frac{\partial}{\partial \varphi} \left(\frac{1}{\sqrt{2\pi}} e^{im\varphi} \right) \\ &= \frac{m\hbar}{\sqrt{2\pi}} e^{im\varphi} = m\hbar \psi_m \end{aligned}$$

$\rightarrow$ ψ 是 L_z 的本征函数, $m\hbar$ 是 L_z 的本征值.

$$(4): \text{令 } u = \cos \theta, \quad \psi(\theta) = P(u)$$

$$\frac{d}{du} \left[(1-u^2) \frac{dP}{du} \right] + \left(\alpha - \frac{m^2}{1-u^2} \right) P = 0$$

$$\text{结果: } \alpha = l(l+1), \quad |m| \leq l$$

$$\hat{L}^2 Y_{l,m} = \hat{L}^2 Y_{l,m} = l(l+1)\hbar^2 Y_{l,m}$$

$$\rightarrow L = l(l+1)\hbar^2$$

$$\text{解: } P(u) = P_l^{(m)}(u)$$

$$\begin{aligned} \text{关联勒让德多项式} &= (1-u^2)^{\frac{|m|}{2}} \frac{d^{|m|}}{du^{|m|}} P_l(u) \\ &= (1-u^2)^{\frac{|m|}{2}} \frac{d^{|m|}}{du^{|m|}} \left[\frac{1}{2^l l!} \frac{d^l}{du^l} (u^2-1)^l \right] \end{aligned}$$

$$\rightarrow Y_{l,m}(\theta, \varphi) = N_{l,m} P_l^{(m)}(\cos \theta) e^{im\varphi}$$

$$\text{其中归一化常数 } N_{l,m} = \sqrt{\frac{(l-|m|)! (2l+1)}{4\pi (l+|m|)!}}$$

性质: (1) $|Y_{l,m}|^2$ 与 φ 无关 $\rightarrow$ 球对称性

2. 奇点

$$Y_{lm}(\lambda - \theta, \phi + \lambda) = (-1)^l Y_{lm}(\theta, \phi)$$

②: 在静电场中 $V(r) = \frac{l(l+1)\hbar^2}{2m_p r^2} - \frac{Ze^2}{4\pi\epsilon_0 r}$

$$\left[-\frac{\hbar^2}{2m_p r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right) + \frac{l(l+1)\hbar^2}{2m_p r^2} - \frac{Ze^2}{4\pi\epsilon_0 r} \right] R = ER$$

令 $\sigma = \frac{m_p Ze^2}{4\pi\epsilon_0 \hbar^2}$, $\rho = 2kr$

$$\Rightarrow \frac{d^2 R}{d\rho^2} + \frac{2}{\rho} \frac{dR}{d\rho} + \left[\frac{\sigma}{\rho} - \frac{1}{4} - \frac{l(l+1)}{\rho^2} \right] R = 0$$

附录 3B 跃迁的选择规则

$$B_{ji} = \frac{\omega_A^2 e^2}{3\hbar^2} |\langle \vec{r}_{ij} \rangle|^2$$

$$\text{电偶极跃迁: } e \langle \vec{r}_{ij} \rangle = e \langle j | \vec{r} | i \rangle = e \iiint \psi_j^* \vec{r} \psi_i d\tau$$

$$i (n_1, m_1, l_1) \rightarrow j (n_2, l_2, m_2)$$

$$\vec{r} = (r_x, r_y, r_z)$$

$$= (r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta)$$

$$= (r \sin \theta \frac{e^{i\phi} + e^{-i\phi}}{2}, r \sin \theta \frac{e^{i\phi} - e^{-i\phi}}{2i}, r \cos \theta)$$

① ϕ 部分:

$$\int_0^{2\pi} e^{-im_2\phi} \vec{r} e^{im_1\phi} d\phi$$

$$= \left[\frac{1}{2} r \sin \theta \int_0^{2\pi} \left(e^{i(m_1 - m_2 + 1)\phi} + e^{i(m_1 - m_2 - 1)\phi} \right) d\phi, \right.$$

$$\left. \frac{1}{2} r \sin \theta \int_0^{2\pi} \left(e^{i(m_1 - m_2 + 1)\phi} - e^{i(m_1 - m_2 - 1)\phi} \right) d\phi, \right.$$

$$\left. r \cos \theta \int_0^{2\pi} e^{i(m_1 - m_2)\phi} d\phi \right)$$

$$\vec{B} \neq 0$$

$$\Delta m = m_1 - m_2 = 0, \pm 1$$

② θ 部分:

$$\Delta l = \pm 1$$